

Cooper Miles

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Objective

Seeking co-op/internship as a Software Developer. Available: Jan - Dec 2027

Skills

Languages: Python, Java, Javascript, Typescript, HTML/CSS, SQL, Golang, C#, C, GDScript

Technologies: React, Flask, PostgreSQL, Docker, REST APIs, Angular, Gunicorn

Tools: Git, Openshift, Claude, ChatGPT, Fusion360, Blender

Work Experience

Research Co-op/Intern

Rochester Institute of Technology

Rochester, NY

Aug 2026 – Present

- Researched papers on using LLMs for computer forensics for security and privacy of protected healthcare information
- Completed CITI Program training for protecting the security and privacy of human subjects' data
- Investigated areas where LLMs can be used to strengthen HIPAA compliance
- Required project deliverables include a research paper submitted to an international conference in 2027

Projects

Archipelago Hosting Service | May 2026 - Present | github.com/Titanium-Lung/archipelago-csh

Developed a full-stack web application using React and Flask to create and manage multiplayer Archipelago game servers

- Deployed and configured the application on OpenShift, creating application deployments and routes, and connecting the application to a PostgreSQL database
- Designed a PostgreSQL-backed persistence layer for storing active game data and restoring hosted rooms after application restarts, allowing rooms to shut down after inactivity and be restored as needed
- Designed a process-management architecture to coordinate Archipelago server subprocesses across multiple Gunicorn workers, using TCP and JSON messages to communicate with a centralized process manager

Mock Ufund Website | Aug - Dec 2025

Developed a full-stack web application with a Java REST API backend and Angular frontend as part of a four-person team

- Implemented a pet adoption system for users to submit applications and administrators to approve or deny
- Used interfaces and polymorphism to decouple the REST controller from the underlying data-access implementation, improving maintainability and extensibility
- Collaborated through Git to integrate backend and frontend components and coordinate development across the team

Basketpong | Jan - Mar 2025 | github.com/Titanium-Lung/basketpong

Developed a 2D multiplayer game in Godot/GDScript, combining basketball-style gameplay with Pong mechanics

- Implemented collision detection, ball physics, player movement, and scoring mechanics to create responsive gameplay
- Designed and implemented a multi-page user interface for navigating menus

Education

Rochester Institute of Technology, Rochester, New York

Bachelor of Science in Computer Science. GPA: 3.80, Dean's List since Fall 2024

Expected Graduation May 2029

- **Relevant courses:** Intro to AI, Mechanics of Programming, Analysis of Algorithms, Intro to Software Engineering

Activities

House History Director and Executive Board Member

Computer Science House

Jun 2026 – Present

- Supervised house history operations, and maintained, upheld, and promoted house traditions
- Interacted with companies to select a venue for house's annual formal dinner
- Presented issues to E-Board for discussion and approval